
XorOJ

A Competitive Programming Online Judge Platform

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What is XorOJ?

- A full-stack online judge platform for competitive programming
- Supports problem creation, contests, leaderboards, and user analytics
- Inspired by platforms like Codeforces and DMOJ



Key Idea

Submit code. Get instant verdicts.

*Compete in contests.
Climb the leaderboard.*

Platform Features

User System

- Registration & JWT login
- Profile with bio, institute, country
- Submission & contest history
- Role-based access control

Problem System

- Create/edit problems (Markdown)
- Difficulty ratings (800–4000)
- Tags, test cases, validators
- Custom checkers & generators
- Problem contributors

Contest System

- Create & manage contests
- ICPC-style standings
- Contest registration
- Per-contest submissions
- Standings snapshots

Platform Features

Code Execution

- GCC 13
- Time & memory limit enforcement
- 8 verdict types (AC, WA, TLE...)

Analytics & Dashboard

- Global leaderboard (DENSE_RANK)
- Activity heatmap (GitHub-style)
- Submission analytics by verdict
- Contest performance summary
- Platform-wide statistics

Advanced Features

- Tag-based problem recommendations
- Blog system (create/view/list)
- Audit trail logging
- Math rendering (MathJax)
- In-browser code editor (Monaco)

Tools & Technology

Backend

Java 24

Spring Boot 3.5

Spring Security

Spring Data JPA

Hibernate 6

JWT (jjwt)

Lombok

Maven

Frontend

React 19

Vite 7

Tailwind CSS

DaisyUI

MUI

Axios

React Router

MathJax

Monaco Editor

Database

PostgreSQL 14+

Infrastructure

Local File Storage

Dev Tools

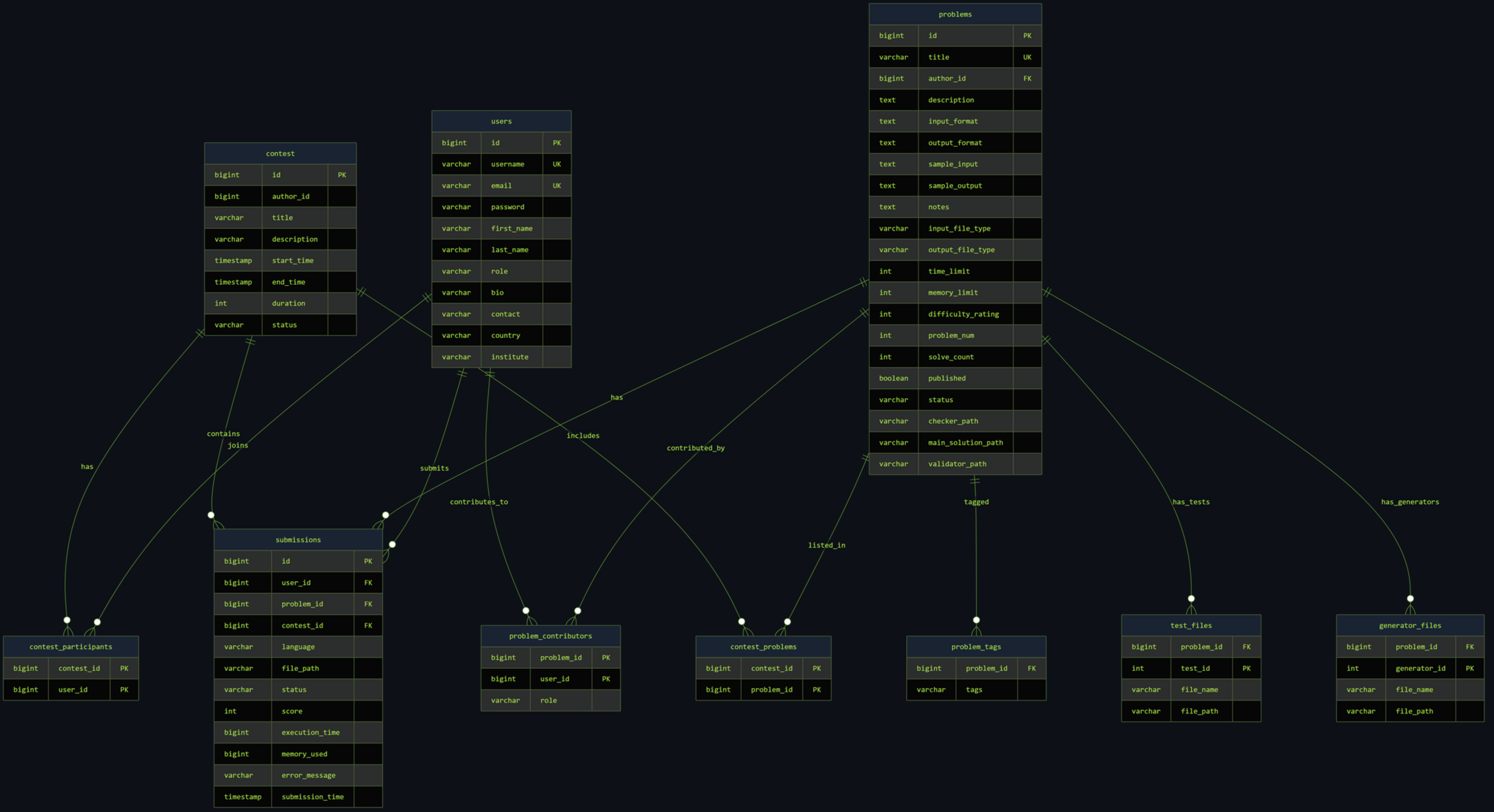
Git & GitHub

VS Code

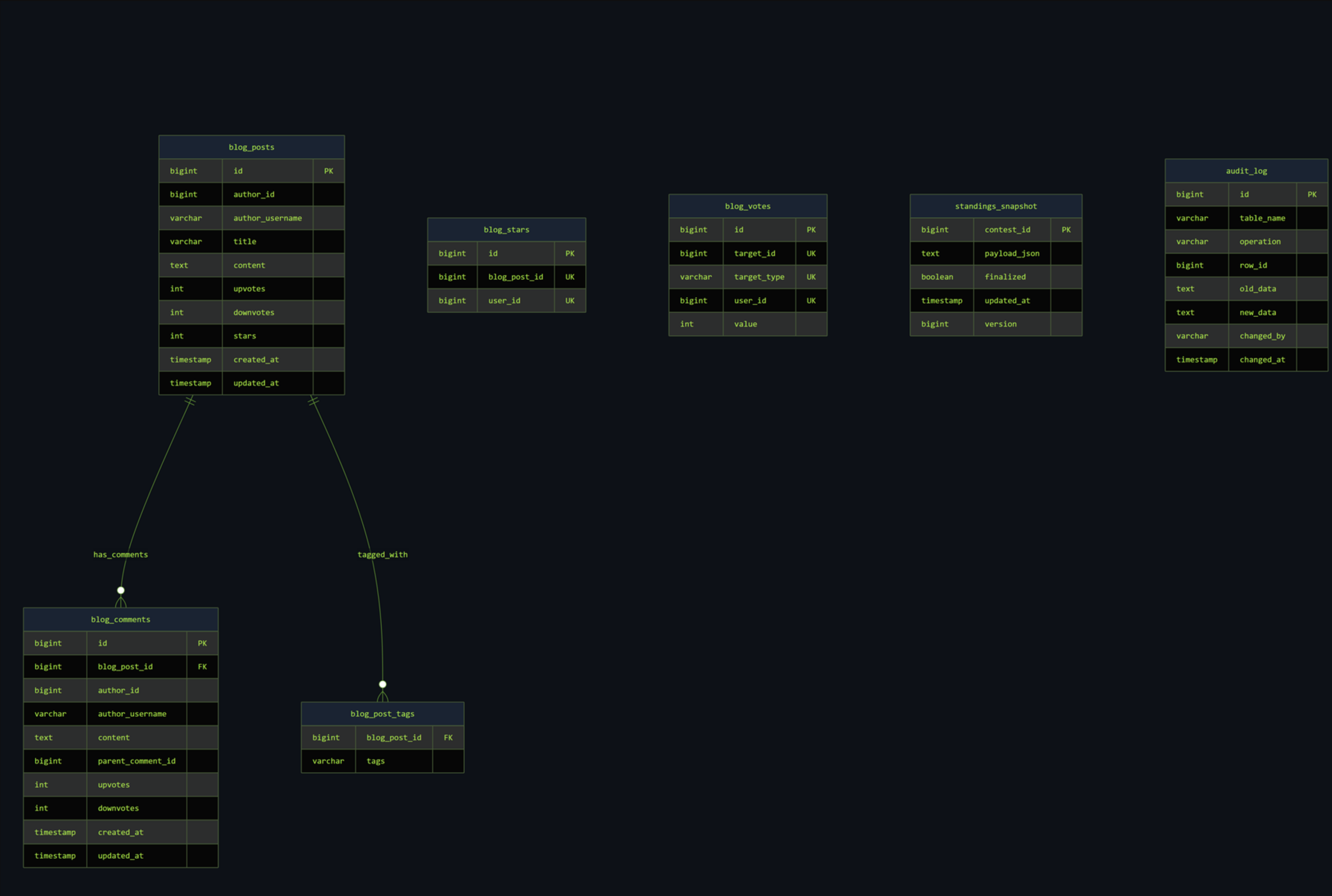
IntelliJ IDEA

Postman

Database Design — ER Diagram



Database Design — ER Diagram



Database — Core Tables

TABLE	PURPOSE
users	User accounts, auth, profiles
problems	Problem statements, limits, settings
submissions	Code submissions & verdicts
contest	Contest metadata & schedule
contest_problems	M:N — contest ↔ problems
contest_participants	M:N — contest ↔ users
test_files	Test case I/O files per problem
generator_files	Test case generators per problem
standings_snapshot	ICPC standings JSON cache
audit_log	Change tracking (JSONB diff)

RELATIONSHIP	TYPE
users → submissions	1:N
problems → submissions	1:N
contest → submissions	1:N
contest ↔ problems	M:N
contest ↔ users	M:N
problems → test_files	1:N
problems → tags	1:N
problems → contributors	1:N

ORM
• Hibernate 6 + JPA • DDL: hibernate.ddl-auto=update • Manual SQL for triggers, views, functions

Advanced PostgreSQL Features

Triggers (5)

- Audit trail on users, problems, submissions, contest
- Auto-update solve_count on submissions change

Stored Functions (6)

- Global leaderboard (DENSE_RANK window fn)
- User submission analytics (cursor)
- Activity heatmap (generate_series)
- Tag-based recommendations (CTEs + cursor)
- Platform dashboard statistics
- Contest performance

Materialized View

- mv_user_statistics — precomputed user stats
- Concurrent refresh, debounced to 30s

Indexes (12+)

- Composite indexes for common queries
- Partial indexes (e.g., ACCEPTED only)
- Covering indexes for index-only scans
- Unique index on MV for concurrent refresh

Cursors (3)

Submission analytics · Recommendations
· Contest performance

Codebase Architecture

Backend (Spring Boot)

- 12 REST Controllers
 - Auth, Problem, Contest, Submission
 - Profile, Blog, Standings, AdvancedDB
 - ProblemEditor, ContestEditor, JudgePanel
- 14 Service classes
 - JudgingService — Docker-based execution
 - ScoreboardService — ICPC standings
 - AdvancedDBService — DB analytics
 - FileStorageService — local file mgmt
- JWT authentication & Spring Security
- Multipart file upload (10MB limit)

Frontend (React + Vite)

- 27+ Page components
 - Home, ProblemSet, ProblemPage
 - ContestList, ContestView, Standings
 - Profile, Analytics, Leaderboard
 - Blog, Dashboard, Recommendations
 - Problem/Contest Editors
- 10 Shared components
 - CodeEditor, IDE (Monaco-based)
 - Header, Footer, Card, CountdownTimer
 - MathRenderer (MathJax)

User Flow & Demo



Problem Author Flow

- Create problem with Markdown editor
- Upload test cases, validators, checkers
- Set time/memory limits & difficulty
- Add to contests

Contest Flow

- Create contest with start/end times
- Attach problems to contest
- Users register and compete
- Live ICPC-style standings

CONTRIBUTIONS

Scan for Github



DEMO

Thank You

Questions?